

# Exercises

Category Theory, M1 MPRI, ENS Paris-Saclay

First period, 2026-27

## 1 Preorders and monoids as categories

**Exercise 1** (Preorders as thin categories). A category  $C$  is *thin* if for any objects  $a, b$  of  $C$ , there is at most one morphism from  $a$  to  $b$ . A *preorder* is a set  $P$  equipped with a binary relation  $\leq$  which is reflexive and transitive.

Prove that a small thin category is essentially the same thing as a preordered set. More precisely, define mutually inverse assignments between the collection of small thin categories and the collection of preorders.

**Exercise 2** (Monoids as one-object categories). Prove that a small category with one object is essentially the same thing as a monoid.

## 2 Isos, epis, monos

**Definition A.** Let  $f: a \rightarrow b$  be a morphism in a category  $C$ . For any object  $c$  of  $C$ , we define a function  $f_c^*: C(b, c) \rightarrow C(a, c)$  by  $f_c^*(h) := hf$  for any  $h: b \rightarrow c$ . We also define a function  $f_c^+: C(c, a) \rightarrow C(c, b)$  by  $f_c^+(g) := fg$  for any  $g: c \rightarrow a$ .

**Exercise 3** (Isomorphisms in a category). Let  $f: a \rightarrow b$  be a morphism in a category  $C$ . Prove that the following four properties are equivalent:

- (i)  $f$  is an isomorphism, i.e., there exists  $g: b \rightarrow a$  such that  $fg = \text{id}_b$  and  $gf = \text{id}_a$ ;
- (ii)  $f$  has a left inverse and a right inverse, i.e., there exist  $p: b \rightarrow a$  and  $s: b \rightarrow a$  such that  $fs = \text{id}_b$  and  $pf = \text{id}_a$ ;
- (iii) for any object  $c$  of  $C$ , the function  $f_c^*$  is a bijection;
- (iv) for any object  $c$  of  $C$ , the function  $f_c^+$  is a bijection.

**Exercise 4** ( $\text{Iso} \subsetneq \text{Mono} \cap \text{Epi}$ ). This exercise shows that the class of isomorphisms is included in the class of morphisms that are both mono and epi, and that this inclusion is strict in general.

- a. Prove that an isomorphism is both a monomorphism and an epimorphism.
- b. Prove that, in the category **Mon** of monoids and homomorphisms, the inclusion function  $i: \mathbb{N} \hookrightarrow \mathbb{Z}$  is both a monomorphism and an epimorphism, but not an isomorphism.

*Hint.* Recall that a monoid homomorphism must preserve any existing inverses.

**Exercise 5** (Split epimorphisms). Let  $C$  be a category. A morphism  $p: a \rightarrow b$  in  $C$  is called a *split epimorphism* if it has a right inverse, i.e., if there exists  $s: b \rightarrow a$  such that  $ps = \text{id}_b$ .

- Prove that a split epimorphism is an epimorphism.
- Give an example of a category  $C$  in which not every epimorphism is a split epimorphism.
- Prove that  $p$  is a split epimorphism if, and only if, for any object  $c$  of  $C$ , the function  $p_c^+$  (see Definition A) is surjective.
- Show, without using the Axiom of Choice, that a function is surjective if, and only if, it is an epimorphism in the category **Set**.
- Recall that the Axiom of Choice (AC) states that, for any set  $\mathcal{A}$  of non-empty sets, there exists a function  $f$  with domain  $\mathcal{A}$  and such that, for every  $A \in \mathcal{A}$ ,  $f(A) \in A$ . Prove that AC is equivalent to: in the category **Set**, every epimorphism is a split epimorphism.

### 3 Functors

**Exercise 6.** Let  $C$  be the category whose objects are the vector spaces  $\mathbb{R}^d$  for each  $d \in \mathbb{N}$ , and whose morphisms are the linear transformations. Let  $\mathbf{Mat}_{\mathbb{R}}$  be the category whose objects are natural numbers, whose morphisms from  $k$  to  $n$  are the  $n \times k$  real matrices, and whose composition is given by matrix multiplication.

Define a functor  $F$  from  $\mathbf{Mat}_{\mathbb{R}}$  to  $C$  which is an *isomorphism*, i.e.,  $F$  is a bijection on objects and, for any objects  $n, k$  of  $\mathbf{Mat}_{\mathbb{R}}$ , the function  $F: \mathbf{Mat}_{\mathbb{R}}(k, n) \rightarrow C(Fk, Fn)$  is a bijection.

**Exercise 7.** Let  $\Sigma$  be a set. Using Exercise 2, regard the monoid  $\Sigma^*$  as a one-object category. Denote by  $\mathbf{Set}_f$  the category of finite sets and functions between them. Prove that the collection of functors  $F: \Sigma^* \rightarrow \mathbf{Set}_f$  is in bijection with the collection of pairs  $(Q, \delta)$ , where  $Q$  is a finite set and  $\delta: \Sigma \times Q \rightarrow Q$  is a function.

**Exercise 8.** Let  $F: C \rightarrow D$  be a functor.

- Prove that  $F$  preserves isomorphisms, i.e., if  $f$  is an iso in  $C$ , then  $Ff$  is an iso in  $D$ .
- Show that  $F$  does not necessarily preserve monomorphisms or epimorphisms.

**Exercise 9.** A *quiver* is a tuple  $G = (V, E, d, c)$ , where  $V$  and  $E$  are sets, and  $d, c: E \rightrightarrows V$  are functions. For  $a, b \in V$ , a *path* from  $a$  to  $b$  in  $G$ , is a list  $p = (a_n, f_n, a_{n-1}, f_{n-1}, \dots, a_1, f_1, a_0)$ , for some  $n \geq 0$ , with  $a_i \in V$  and  $f_i \in E$  for each  $0 \leq i \leq n$ , such that  $a_0 = a$ ,  $a_n = b$ , and, for each  $0 < i \leq n$ ,  $d(f_i) = a_{i-1}$  and  $c(f_i) = a_i$ ; we write  $p: a \rightarrow b$ . The *category of paths* on  $G$ ,  $PG$ , has set of objects  $V$  and set of morphisms the paths in  $G$ .

- Define an appropriate identity and composition on  $PG$  and show that  $PG$  is a category.
- Let  $C$  be a category. A *morphism* from  $G$  to  $C$  is a pair of functions  $F_0: V \rightarrow \text{ob } C$  and  $F_1: E \rightarrow \text{mor } C$  such that, for all  $f \in E$ ,  $F_1 f$  has domain  $F_0(df)$  and codomain  $F_0(cf)$ . Prove that, for any morphism  $(F_0, F_1)$  from  $G$  to  $C$ , there is a unique functor  $\bar{F}: PG \rightarrow C$  such that  $\bar{F}a = F_0a$  for all  $a \in V$  and  $\bar{F}(ce, e, de) = F_1e$  for all  $e \in E$ .

*Remark.* The category  $PG$  is also called the *free category* on  $G$ .

## 4 Binary products and terminal objects

**Exercise 10.** Let  $P$  be a preorder, seen as a category (see Exercise 1).

- a. Let  $t \in P$ . Show that  $t$  is terminal if, and only if,  $t$  is maximal.
- b. Let  $a, b, c \in P$ . Show that  $c$  is a product of  $a$  and  $b$  if, and only if,  $c$  is a greatest lower bound (infimum) of  $a$  and  $b$ .

**Exercise 11.** a. Give an example of sets  $A$  and  $B$  such that the projection  $A \times B \rightarrow B$  is not surjective.

- b. Conclude that product projections need not be epimorphisms.
- c. Show that the unique map to a terminal object need not be an epimorphism.

**Definition B.** Let  $I$  be an index set,  $C$  a category,  $a$  an object, and  $(f_i: a \rightarrow b_i)_{i \in I}$  a family of morphisms with domain  $a$ . We say that the family  $(f_i)_{i \in I}$  is *jointly monomorphic* if, for any pair of morphisms  $g, h: x \rightrightarrows a$ , if  $f_i \circ g = f_i \circ h$  for every  $i \in I$ , then  $g = h$ .

We say that a jointly monomorphic family  $(f_i: a \rightarrow b)_{i \in I}$  is *extremal* if, for any family of morphisms  $(f'_i: a' \rightarrow b_i)_{i \in I}$  and any epimorphism  $e: a \rightarrow a'$ , if  $f_i = f'_i \circ e$  for every  $i \in I$ , then  $e$  is an isomorphism.

**Exercise 12.** Let  $C$  be a category, let  $b_1, b_2$  be objects in  $C$  and let  $p_1: b_1 \times b_2 \rightarrow b_1$ ,  $p_2: b_1 \times b_2 \rightarrow b_2$  be a product.

- a. Prove that the family  $(p_1, p_2)$  is jointly monomorphic and extremal.
- b. Let  $(f_i: a \rightarrow b_i)_{i=1,2}$  be a family of morphisms with domain  $a$ . Prove that the family is jointly monomorphic if, and only if,  $\langle f_1, f_2 \rangle$  is a monomorphism.

## 5 Binary coproducts and initial objects

**Definition C.** Let  $C$  be a category and  $a_1, a_2$  objects of  $C$ . A pair of morphisms  $i_1: a_1 \rightarrow b$  and  $i_2: a_2 \rightarrow b$  is a *coproduct* if it is a product in  $C^{\text{op}}$ . An object  $a$  of  $C$  is *initial* if it is terminal in  $C^{\text{op}}$ .

**Exercise 13.** a. Write down a more concrete definition of ‘coproduct’ and of ‘initial’.

- b. Explain why coproducts and initial objects are unique up to isomorphism.
- c. Show that coproducts and an initial object exist in **Set**.
- d. What are a coproduct and initial object in a preorder?

**Notation.** A coproduct of two objects  $a_1, a_2$  is denoted  $a_1 + a_2$ , and an initial object is denoted  $0$ . The copairing of two morphisms  $f_1: a_1 \rightarrow c$  and  $f_2: a_2 \rightarrow c$  is denoted  $[f_1, f_2]: a_1 + a_2 \rightarrow c$ . The functions  $i_k: a_k \rightarrow a_1 + a_2$  are called *coprojections*, or also *injections* or *inclusions* (note that they are not necessarily monomorphisms, by the dual of Exercise 11).

**Exercise 14.** Let  $C$  be a category in which any two objects have a product and a coproduct.

- a. Show that, for any objects  $a, b, c$ , there exists a morphism  $(a \times b) + (a \times c) \rightarrow a \times (b + c)$ .
- b. Does there always exist an isomorphism?

**Exercise 15.** Consider the monoids  $(\mathbb{N}, 0, +)$  and  $(\{a, b\}^*, \epsilon, \cdot)$ .

- a. Prove that  $\{a, b\}^*$  is a coproduct of  $\mathbb{N}$  and  $\mathbb{N}$  in the category **Mon**.
- b. What is the coproduct of  $\{a, b\}^*$  and  $\mathbb{N}$  in **Mon**?
- c. What is the coproduct of  $\mathbb{Z}$  and  $\mathbb{Z}$  in the category of groups?

**Exercise 16.** Let  $\Sigma$  be a finite set. Consider the following category, which we denote  $\Sigma/\mathbf{Mon}$ :

- objects are pairs  $(M, f)$ , where  $M$  is a monoid, and  $f: \Sigma \rightarrow M$  is a function from  $\Sigma$  to (the underlying set of)  $M$ ;
- a morphism  $(M, f) \rightarrow (N, g)$  is a monoid homomorphism  $h: M \rightarrow N$  such that  $h \circ f = g$ .

Show that  $\Sigma/\mathbf{Mon}$  has an initial object.

**Exercise 17.** Let  $C$  be a category with a terminal object  $1$  and an initial object  $0$ . Prove that any morphism  $f: 1 \rightarrow 0$  is an isomorphism.

## 6 Equalizers and coequalizers

**Exercise 18.** Prove that any equalizer is a monomorphism.

**Exercise 19.** Let  $f, g: A \rightrightarrows B$  be two functions in **Set**. Define  $\sim$  to be the smallest equivalence relation on  $B$  such that  $f(a) \sim g(a)$  for every  $a \in A$ . Prove that the quotient map  $e: B \rightarrow B/\sim$  is a coequalizer of  $f$  and  $g$ .

## 7 Pullbacks and pushouts

**Definition D.** The *walking span*,  $\Lambda$ , is the (poset) category with three objects and two non-identity morphisms as depicted on the left below. The *walking cospan*,  $V$ , is its opposite category, as depicted on the right below.



A *span* in a category  $C$  is a diagram of shape  $\Lambda$ , and a *cospan* is a diagram of shape  $V$ .

A *pushout*<sup>1</sup> is a colimit of a span, and a *pullback*<sup>2</sup> is a limit of a cospan.

**Exercise 20.** a. Give more explicit definitions of pushout and pullback.

<sup>1</sup>In French: *somme amalgamée*.

<sup>2</sup>In French: *produit fibré*.

- b. Let  $f_1: b_1 \rightarrow a$  and  $f_2: b_2 \rightarrow a$  be morphisms in a category  $C$ . Suppose that a product  $b_1 \times b_2$  of  $b_1$  and  $b_2$  exists in  $C$ , with projections  $p_i: b_1 \times b_2 \rightarrow b_i$  for  $i = 1, 2$ , and that the pair of morphisms  $f_1 p_1, f_2 p_2: b_1 \times b_2 \rightrightarrows a$  has an equalizer,  $e: c \rightarrow b_1 \times b_2$ . Prove that  $(p_1 e, p_2 e)$  is a pullback of  $(f_1, f_2)$ .
- c. Let  $f_1: B_1 \rightarrow A$  and  $f_2: B_2 \rightarrow A$  be a cospan in **Set**. Show that there exists a pullback with summit  $P := \{(b_1, b_2) \in B_1 \times B_2 \mid f_1(b_1) = f_2(b_2)\}$ .
- d. Let  $f_1: A \rightarrow B_1$  and  $f_2: A \rightarrow B_2$  be a span in **Set**. Compute its pushout.

**Exercise 21.** Let  $C$  be a category containing the following two commuting squares.

$$\begin{array}{ccccc} a & \xrightarrow{p} & b & \xrightarrow{t} & e \\ \downarrow q & & \downarrow r & & \downarrow v \\ c & \xrightarrow{s} & d & \xrightarrow{u} & f \end{array}$$

Suppose that  $(r, t)$  is a pullback of  $(u, v)$ . Prove that  $(tp, q)$  is a pullback of  $(v, us)$  if, and only if,  $(p, q)$  is a pullback of  $(r, s)$ .

**Exercise 22.** Let  $C$  be a category in which pullbacks and a terminal object exist.

- a. Prove that binary products exist in  $C$ .
- b. Prove that all equalizers exist in  $C$ .

**Exercise 23.** Prove that a morphism  $f: a \rightarrow b$  is mono if, and only if, the square

$$\begin{array}{ccc} a & \xrightarrow{\text{id}_a} & a \\ \downarrow \text{id}_a & & \downarrow f \\ a & \xrightarrow{f} & b \end{array}$$

is a pullback.

## 8 Limits in general

**Exercise 24.** The aim of this exercise is to prove that *a category with small products and equalizers is complete*.

Let  $C$  be a category and let  $D: I \rightarrow C$  be a diagram in  $C$ , with  $I$  small. For each morphism  $t$  in  $I$ , write  $dt$  for its domain and  $ct$  for its codomain.

Suppose that the family of objects  $(D_i)_{i \in \text{ob } I}$  has a product  $(p_i: P \rightarrow D_i)_{i \in \text{ob } I}$  in  $C$ , and that the family of objects  $(D_{ct})_{t \in \text{mor } I}$  has a product  $(q_t: Q \rightarrow D_{ct})_{t \in \text{mor } I}$  in  $C$ .

Consider the morphisms  $f, g: P \rightrightarrows Q$  defined as

$$f := \langle p_{ct} \rangle_{t \in \text{mor } I} \text{ and } g := \langle Dt \circ p_{dt} \rangle_{t \in \text{mor } I},$$

that is,  $q_t \circ f = p_{ct}$  and  $q_t \circ g = Dt \circ p_{dt}$  for every  $t \in \text{mor } I$ .

Suppose that  $e: E \rightarrow P$  is an equalizer of the pair  $(f, g)$ .

Prove that  $(p_i \circ e: E \rightarrow D_i)_{i \in \text{ob } I}$  is a limit cone for  $D$ .

**Definition E.** A poset  $(I, \leq)$  is *down-directed* if  $I$  is non-empty and, for any  $a, b \in I$ , there exists  $c \in I$  such that  $c \leq a$  and  $c \leq b$ . A diagram  $D: I \rightarrow C$  is *projective* if  $I$  is a down-directed poset. A category  $C$  has *projective limits* if any projective diagram in  $C$  has a limit.

**Exercise 25.** The aim of this exercise is to prove that *a category with finite products and projective limits has all small products*. Combined with the previous exercise, it follows that *a category with finite and projective limits is complete*.

Let  $(a_j)_{j \in J}$  be a family of objects in a category  $C$ . Define  $I$  to be the set of finite subsets of  $J$ , ordered by reverse inclusion, i.e., for  $S, T \in I$ ,  $S \leq T$  if, and only if  $T \subseteq S$ .

- a. Prove that  $(I, \leq)$  is down-directed.

For each  $S$  in  $I$ , suppose that  $(p_i^S: D_S \rightarrow a_i)_{i \in S}$  is a product of  $(a_i)_{i \in S}$  in  $C$ .

- b. Show that whenever  $S, T \in I$  and  $S \leq T$ , there exists a unique morphism  $D_{ST}: D_S \rightarrow D_T$  in  $C$  such that, for any  $i \in T$ , we have  $p_i^T \circ D_{ST} = p_i^S$ .
- c. Conclude that  $D: I \rightarrow C$  is a diagram.

Suppose that  $(q_S: P \rightarrow D_S)_{S \in I}$  is a limit of  $D$ .

- d. Show that  $(p_j^{\{J\}} \circ q_{\{j\}}: P \rightarrow a_j)_{j \in J}$  is a product of the family  $(a_j)_{j \in J}$ .

**Exercise 26.** Let  $C$  be a small category in which all small families have products. Prove that  $C$  is thin and complete.